

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Schedule of Crossings
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-LST-002
Document Revision	A01
Document Status	Issued For Approval
Originator	E. Uwaoma
Security Classification	Restricted
Issue Date	22 Jul 2022

Schedule of Crossings

A01	22.07.22	Issued for Approval	E.Uwaoma	E.Nwachukwu	A.Duru	
R01	08.06.22	Issued for Review	E.Uwaoma	E.Nwachukwu	A.Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	08/06/2022	Issued for Review
A01	22/07/2022	Issued for Approval

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-LST-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

	Document No.	EA-BFPCL-GPMC-GGPL-FEED-PPL-LST-002	REV	A01	
	Doc. Title	Schedule of Crossings			

Purpose of Document:

This document present the list of all crossings which straddle along the entire right of way of the Gas Gathering Pipelines & Associated Infrastructure- Phase 1

Reference:

(1) EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 038

(2) BFPCL Bubouwe - Akabel Topographical Survey Draft Report_Rev.2_08022022

The Pipeline system comprises of two segments.

(1) 24" x 6.7 Km pipeline system departing the BOUBUWE (BBOU) Manifold Station to ELEPA (ELEP) manifold station.

(2) 36" X 31.5" Km pipeline system departing the ELEPA manifold station to AKEBEL (AKAB)

Table of Contents

Crossing List	6
---------------	---

BUBOUWE MANIFOLD STATION TO ELEPA MANIFOLD STATION					
S/N	START KP (Km)	END KP (Km)	CROSSING	LENGTH OF CROSSING (m)	PROTECTION TYPE
1	0+450	0+780	RIVER	330	HDD
2	1+560	1+600	CREEK	40	CONCRETE COATING
3	2+060	2+120	RIVER	60	CONCRETE COATING
4	3+560	3+600	CREEK	40	CONCRETE COATING
5	4+040	4+080	CREEK	40	CONCRETE COATING
6	4+200	4+480	RIVER	280	HDD
7	5+440	5+520	CREEK	80	CONCRETE COATING
8	5+700	6+080	RIVER	380	HDD
9	6+660	6+700	CREEK	40	CONCRETE COATING
ELEPA MANIFOLD STATION TO AKABEL PIGGING STATION					
S/N	START KP (Km)	END KP (Km)	CROSSING	LENGTH OF CROSSING (m)	PROTECTION TYPE
1	0+520	0+600	CREEK	80	CONCRETE COATING
2	1+100	1+260	CREEK	60	CONCRETE COATING
3	2+100	2+600	CREEK	500	CONCRETE COATING
4	3+000	3+200	CREEK	200	CONCRETE COATING
5	3+340	3+420	CREEK	80	CONCRETE COATING
6	3+780	3+920	RIVER	140	CONCRETE COATING
7	4+240	4+300	CREEK	60	CONCRETE COATING
8	5+020	5+080	CREEK	60	CONCRETE COATING
9	6+260	6+320	CREEK	60	CONCRETE COATING
10	6+360	6+480	CREEK	120	CONCRETE COATING
11	6+920	6+960	CREEK	40	CONCRETE COATING
12	8+300	9+500	RIVER	1200	HDD
13	10+246	10+266	TEBIDADA - BRASS TRUNK PIPELINE	20	
14	11+160	12+260	RIVER	1100	HDD
15	13+780	13+860	CREEK	80	CONCRETE COATING
16	13+879	13+902	SHELL AGIP PIPELINE	20	
17	15+454	15+474	SHELL AGIP PIPELINE	20	
18	15+840	16+900	RIVER	1060	HDD
19	17+560	17+720	CREEK	160	CONCRETE COATING
20	18+160	18+240	RIVER	80	CONCRETE COATING
21	18+960	19+000	CREEK	40	CONCRETE COATING
22	19+500	19+720	RIVER	220	HDD
23	20+560	20+640	RIVER	100	CONCRETE COATING
24	20+870	20+920	OGADA - BRASS PIPELINE	50	
25	21+160	21+280	CREEK	120	CONCRETE COATING
26	21+400	21+680	CREEK	280	CONCRETE COATING

27	22+100	22+700	RIVER	600	HDD
28	23+000	23+220	CREEK	220	CONCRETE COATING
29	24+900	25+140	CREEK	240	CONCRETE COATING
30	25+300	23+880	CREEK	380	CONCRETE COATING
31	26+080	26+140	CLEARED / PROPOSED NOAC PIPELINE ROW	60	
32	26+820	26+940	CREEK	120	CONCRETE COATING
33	28+440	28+800	RIVER	360	HDD
34	30+400	30+720	CREEK	320	CONCRETE COATING
35	31+040	31+120	CREEK	80	CONCRETE COATING